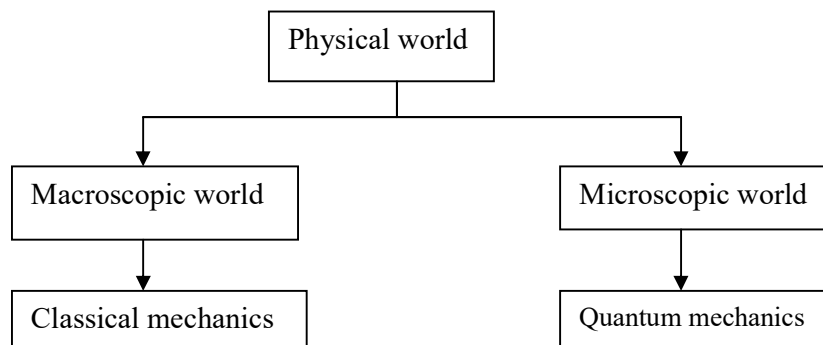


UNIT III

QUANTUM MECHANICS

Introduction:



In the late seventeenth century, Isaac Newton discovered classical mechanics, the law of motion of macroscopic objects.

In the early twentieth century, physicists found that classical mechanics does not correctly describe the behavior of very small particles such as the electrons nuclei (protons & neutrons etc.) and molecules. The behavior of such particles is described by a set of laws called **Quantum mechanics**.

I.e. Classical mechanics applies only to macroscopic particles. For microscopic “particles”. We require a new form of mechanics, called **Quantum mechanics**.

Classical mechanics:

- Classical mechanics deals with the motion and behavior of **macroscopic particles**.
Ex: Motion of planets.
- Classical mechanics is expressed in the language of un-probabilities, certainties.

Quantum mechanics:

- Quantum mechanics deals with the motion and behavior of **microscopic particles**
Ex: Motion of Electrons
- Quantum mechanics is expressed in the language of probabilities, not certain.

DUAL NATURE OF MATTER

de-Broglie hypothesis:

According to de-Broglie hypothesis, every moving particle has always associated with a wave. The particle is controlled by this wave in every aspect.

The wave associated with moving particle of matter (like electrons, protons etc) are known as matter wave or pilot waves or de-Broglie waves.



Figure .1 Motion of particle (Electron)

If a particle of mass m moving with velocity v , then the wavelength of matter wave is given by

$$\lambda = \frac{h}{mv}$$

Where

$h \rightarrow$ Planck's constant,

$m \rightarrow$ mass of particle,

$v \rightarrow$ velocity of the particle

Derivation of de – Broglie's wave length:

According to Planck's quantum theory of radiation (light) the energy of photon is given by

$$E = h\nu \longrightarrow (1)$$

Where $h \rightarrow$ Planck's constant,

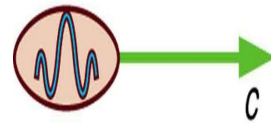
$\nu \rightarrow$ frequency of radiation

According to Einstein's energy-mass relation, the energy of photon is given by

$$E = mc^2 \longrightarrow (2)$$

Where $m \rightarrow$ mass of photon

$C \rightarrow$ velocity of light or photon



F

Figure - Motion of photon

From the above equations (1) and (2); we can write

$$h\nu = mc^2 \quad \left[\because C = \nu\lambda \text{ and } \nu = \frac{c}{\lambda} \right]$$

$$\frac{hc}{\lambda} = mc^2$$

$$\frac{h}{\lambda} = mc$$

$$\lambda = \frac{h}{mc} = \frac{h}{p} \longrightarrow (3)$$

Where $p = mc$ is the momentum of photon having mass m and travelling with velocity c .

de - Broglie suggested that equation (3) can be applied both for photos and material particles. If m is mass of the particle and v is the velocity of the particle, then

Momentum (p) = mv

$$\therefore \text{de - Broglie wave length } \lambda = \frac{h}{mv} \longrightarrow (4)$$

This equation represents that the de-Broglie wave length associated with a particle and also known as de -Broglie's wave equation.

OTHER FORMS OF de -Broglie's WAVE LENGTH:

1. de -Broglie's wave length in terms of energy

If m is mass of the particle and v is the velocity of the particle, then the kinetic energy of the particle;

$$E = \frac{1}{2}mv^2$$

Multiplying by 'm' on both sides; we get

$$m E = \frac{1}{2}m^2v^2$$

$$2 m E = m^2v^2$$

$$(\text{Or}) m v = \sqrt{2 m E}$$

$$\therefore \text{de - Broglie wave length } \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2 m E}} \longrightarrow (5)$$

2. de -Broglie's wave length in terms of voltage

If a charged particle of charge 'q' is accelerated through a potential difference 'v', then

$$E = q v \longrightarrow (6)$$

$$\text{Also we know that the K.E of the particle } E = \frac{1}{2}mv^2 \longrightarrow (7)$$

Equating equations (7) and (6) we get

$$\frac{1}{2}mv^2 = q v$$

Multiplying by 'm' on both sides we get

$$\frac{1}{2}m^2v^2 = m q v$$

$$\therefore m^2v^2 = 2 m q v$$

$$m v = \sqrt{2 m q v} \longrightarrow (8)$$

Substituting equation 8 in equation 4 , we get

$$\therefore \text{de - Broglie wave length } \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mqV}} \longrightarrow (9)$$

3.de –Broglie’s wave length for electron

Substituting the values of m, q and h in equation 9 we get

$$\begin{aligned} \lambda &= \frac{6.626 \times 10^{-34} \text{ JS}}{\sqrt{2 \times 9.1 \times 10^{-31} \text{ Kg} \times 1.6 \times 10^{-19} \text{ CXV}}} \\ &= \frac{12.27 \times 10^{-1} \text{ m}}{\sqrt{V}} \quad [\because 1 \text{ A}^0 = 10^{-10} \text{ m}] \\ \lambda &= \frac{12.27}{\sqrt{V}} \text{ A}^0 \end{aligned}$$

This equation represents the de – Broglie wave length associated with an electron.

4.Wave velocity:

We know that

$$E = h\nu \text{ and } E = mc^2$$

$$h\nu = mc^2$$

$$\nu = \frac{mc^2}{h}$$

The wave velocity is given by

$$\omega = \nu\lambda$$

$$\begin{aligned} \omega &= \frac{mc^2}{h} \cdot \frac{h}{mv} \\ \therefore \omega &= \frac{c^2}{V} \end{aligned}$$

Thus matter waves travel faster than velocity of light.

Properties of de-Broglie waves:

The following are the properties of matter waves.

1. The de-Broglie wavelength of particle of mass (m) moving with a velocity (V) is given by $\lambda = \frac{h}{mv}$

If $v = 0$, $\lambda \rightarrow \infty$ and if $v \rightarrow \infty$, $\lambda \rightarrow 0$.

This implies that matter waves are associated with material particles only if they are in motion

2. Lesser the velocity of the particle, larger the wave length associated with it.

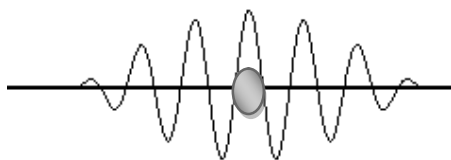
3. Smaller the mass of the particle, the greater the length associated with it.

4. The matter wave of particle is independent of charge (or) nature of the particle.

5. de-Broglie waves are pilot waves. The word, pilot means that these waves pilot (or) guide the particle.

6. Matter waves travel faster than velocity of light, i.e., $\omega = \frac{c^2}{V}$.

7. Matter waves are not electromagnetic waves.
8. The wave nature of matter introduces an uncertainty in the location of the position of the particle because a wave cannot be set exactly at this position (or) at that position.
9. de-Broglie waves are called probability waves. The amplitude of the wave tells us about the probability of finding particle in space at a particular instant. Large wave amplitude means large probability of finding the particle at that position.



NOTE:

1. Mention an application of matter waves?

A. The electron microscope is an application of matter waves, associated with electrons

2. Why cannot we experience the existence of matter waves in our day to day life?

A. $\lambda = \frac{h}{mv}$, if m is large, λ will be small, thus the wave length will be very small, compared to the size of the particle, hence it is not observable.

Heisenberg uncertainty principle:

In 1927, Heisenberg proposed a very interesting principle, which is direct consequence of the dual nature of matter, known as uncertainty principle.

According to Heisenberg uncertainty principle, it is impossible to determine accurately and simultaneously the values of position and momentum of a particle at any time. If Δx is the error (uncertainty) in measuring position, Δp is the error (uncertainty) in measuring momentum; then according to this principle

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Physical significance of wave function:

The wave nature of matter introduced an uncertainty in the location of the position of the particle because a wave cannot be set exactly at this position (or) at that position.

According to Heisenberg uncertainty principle tells us that we cannot detect the particle nature and wave nature at the same time.

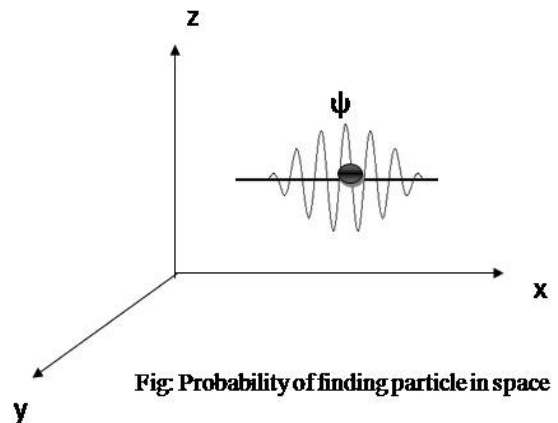
To resolve this uncertainty, Eddington once upon a time humorously suggested that the primordial entity is neither particle nor wave but a 'wavicle'. Thus, de-Broglie waves are called probability waves.

The amplitude of the wave tells us about the probability of finding particle in space at a particular instant. A large wave amplitude means a large probability of finding the particle at that position.

Mathematically, to explain it, scientists introduced a new physical quantity called wave function.

Wave function: - it is a mathematical tool used in quantum mechanics to describe any physical system, and it is denoted by ' ψ '.

- It is variable quantity i.e., associated with a moving particle at any position (x,y,z) and at any time t and it relates probability of a finding the particle at that point and at that time.
- It must be well behaved, that is single valued and continuous everywhere.
- It usually complex quantity and individually it has no meaning.
- To explain it, Max born suggested a new idea about the physical significance of ψ which is generally accepted now a days.
- According Max born $\psi \psi^* = |\psi|^2$ is a real and positive, it has physical meaning. It represents the probability of finding the particle in the state ψ .
- Since the wave function is a complex quantity, it may be expressed in the form of $\Psi_{(x,y,z)} = (a+ib)$



Where a,b are real function of the variables (x,y,z,t) and $i=\sqrt{-1}$.

- The complex conjugate of ψ is given by $\Psi_{(x,y,z)}^* = (a-ib)$

Multiplying the above two equations, we have

$$\Psi \Psi^* = (a+ib)(a-ib) = a^2+b^2$$

According to Max born

$$P = \Psi \Psi^* = |\psi_{(x,y,z)}|^2 = a^2+b^2$$

Thus the product of ψ and ψ^* is real and positive if $\psi \neq 0$ and is known as probability density of the particle associated with the de-Broglie wave

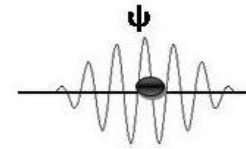


Fig: Probability of finding particle in space

- The probability of finding the particle in a volume $dv = dx \cdot dy \cdot dz$ is given by

$$\text{Probability (p)} = |\psi|^2 dx \, dy \, dz$$

- For the total probability of finding the particle somewhere is (space).

$$P = \iiint |\psi|^2 dx \, dy \, dz = 1$$

- A wave function ψ satisfying the above relation is called a normalized wave function.
- For motion of a particle in one dimensional, the quantity $Pdx = |\psi|^2 dx$ is the probability of finding the particle over a small distance dx at position x .

Schrodinger wave equation:-

Definition: - The equation that describes the wave nature of a particle in mathematical form is known as Schrodinger's wave equation.

There are two types of Schrodinger wave equations they are

- 1) Time independent wave equation and
- 2) Time dependent wave equation

Schrodinger one dimensional Time independent wave equation :-

Consider a particle of mass m moving with velocity v along the x -direction. It is associated with a wave. The displacement of the wave is given by the wave function Ψ . For steady state wave motion, Ψ is a function of x -coordinate only for one dimensional motion. A one dimensional wave equation for the steady wave associated with a particle is given by

$$\psi(x) = Ae^{i(kx - \omega t)} \quad \text{----- (1)}$$

Where, A is the amplitude and λ is wave length of the wave at any point.

Differentiating equation (1) partially with respect to ' x '; we get

$$\frac{d\Psi}{dx} = ikAe^{i(kx - \omega t)}$$

Differentiating once again partially with respect to 'x'; we get

$$\begin{aligned}\frac{d^2\Psi}{dx^2} &= i^2 k^2 A e^{i(kx - \omega t)} \\ &= i^2 k^2 \Psi \quad \text{since, } \psi(x) = A e^{i(kx - \omega t)} \\ &= -k^2 \Psi \quad \text{---- (2) Since, } i^2 = -1\end{aligned}$$

We Know that wave vector $k = 2\pi/\lambda$, Sub the value of k in eqn(2),

$$\begin{aligned}\frac{d^2\Psi}{dx^2} &= -(2\pi/\lambda)^2 \Psi \\ &= -\left(\frac{4\pi^2}{\lambda^2}\right) \Psi \quad \text{----- (2a)}\end{aligned}$$

de-Broglie wavelength associated with the particle is, $\lambda = \frac{h}{mv}$

$$\begin{aligned}\frac{1}{\lambda^2} &= \frac{m^2 v^2}{h^2} \\ &= \frac{2m\left(\frac{1}{2mv^2}\right)}{h^2} \quad \text{----- (3)}\end{aligned}$$

We also know that,

Total energy of a particle = Kinetic energy + Potential energy

$\Rightarrow$ Kinetic energy = Total energy – Potential energy

Hence we can write, K.E as, $\frac{1}{2} mv^2 = (E - v) \quad \text{----- (4)}$

Sub eqn 4 in 3,

$$\frac{1}{\lambda^2} = \frac{2m}{h^2} (E - v) \quad \text{----- (5)}$$

Sub, eqn 5 in 2(a),

$$(d^2\Psi/dx^2) = -(8\pi^2 m/h^2)(E - v)\Psi$$

$$(d^2\Psi/dx^2) + (8\pi^2 m/h^2)(E - v)\Psi = 0$$

$$\text{Or, } \frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E-V)\psi=0$$

Where $\hbar = \frac{h}{2\pi}$ is called plank's reduced constant.

This equation is known as Schrodinger time independent one-dimensional wave equation.

Note:

For 3dimensional:

$$\frac{d^2\psi}{dx^2} + \frac{d^2\psi}{dy^2} + \frac{d^2\psi}{dz^2} + \frac{8\pi^2m}{h^2}(E-V)\psi=0$$

$$\text{Or } \nabla^2 \psi + \frac{8\pi^2m}{h^2}(E-V)\psi=0$$

$\nabla^2 = \frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2}$ is called Laplace operator.

Particle in one dimensional box or well: (Qualitative only)

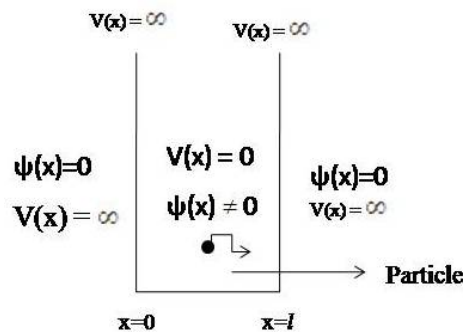


Fig: Particle in one dimensional box

Let us consider a particle of mass 'm' moving with velocity 'v' along x-direction and is confined between two infinite potential rigid walls, so that the particle has no chance of escaping from them. Therefore, the particle bounces back and forth between two walls as shown in fig.

- Let the potential energy of electron inside the box is constant and can be taken as zero for simplicity.
- From fig, we consider two regions :
 - (i) Outside the box
 - (ii) Inside the box

Outside the box:

The particle cannot exist outside and on the walls of the box; therefore the probability of finding the electron outside the wall is zero and the potential energy is ∞ .

i.e.,

$\begin{aligned}\psi(x) &= 0 \\ V(x) &= \infty\end{aligned}$
--

 When $x \leq 0$ and $x \geq l$

Inside the box:

The particle exists inside the box; therefore the probability of finding the electron inside the box is not equal to zero and the potential energy is zero.

i.e.,

$\begin{aligned}\psi(x) &\neq 0 \\ V(x) &= 0\end{aligned}$
--

 When $x > 0$ and $x < l$

- To calculate the probability of finding particle within the box, let us consider one dimensional time independent Schrödinger wave equation

$$\text{i.e., } \frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2} (E - V(x)) \psi = 0 \longrightarrow (1)$$

Inside the box, the potential energy $V(x) = 0$

Therefore, equation (1) becomes

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2mE}{h^2} \psi = 0$$

$$(\text{Or}) \frac{d^2\psi}{dx^2} + k^2\psi = 0 \longrightarrow (2)$$

$$\text{Where } k^2 = \frac{8\pi^2mE}{h^2} \longrightarrow (3)$$

Equation (2) is a second order differential equation; therefore it should have solution with two arbitrary constants.

∴ The solution for equation (2) is given by

$$\Psi(x) = A \sin Kx + B \cos Kx \longrightarrow (4)$$

Where A and B are called as arbitrary constants, which can be found by applying the boundary conditions.

Boundary Condition at $x=0$:

At $x = 0$, the probability of finding the electron is zero, i.e., $\psi(x) = 0$

Equation (4) becomes

$$0 = A \sin K(0) + B \cos K(0)$$

$$0 = 0 + B$$

$\therefore B = 0$

 $\longrightarrow (5)$

Boundary Condition at $x=l$:

At $x = l$, the probability of finding the electron is zero, i.e., $\psi(x) = 0$

Equation (4) becomes

$$0 = A \sin K(l) + B \cos K(l)$$

$$0 = A \sin K(l) + (0) \cos K(l) \quad [\text{Since } B = 0]$$

$$A \sin K(l) = 0$$

$$\text{Since } A \neq 0; \sin K(l) = 0 \longrightarrow (6)$$

$$\text{We know that, } \sin n\pi = 0 \longrightarrow (7)$$

$$\text{Comparing these two equations, we can write } K(l) = n\pi$$

$$K = \frac{n\pi}{l} \longrightarrow (8)$$

$$k^2 = \frac{n^2\pi^2}{l^2} \longrightarrow (9)$$

Substituting the values of 'B' and 'K' in equation (4),
Equation (4) becomes

$$\boxed{\Psi_n(x) = A \sin \frac{n\pi x}{l}} \longrightarrow (10)$$

This equation represents the wave function associated with moving free electron inside the box.

Normalization of the wave function: it is the process by which the probability (P) of finding the particle or electron inside the box can be done.

The total probability that the particle is somewhere in the box must be unity

$$\text{i.e. , } P = \int_0^l |\Psi_n(x)|^2 dx \longrightarrow (11)$$

[Since the particle present inside the box between the length 0 to l, then the limits are chosen between 0 to l]

Substituting equation (10) in equation (11), we get

$$P = \int_0^l A^2 \sin^2 \frac{n\pi x}{l} dx = 1$$

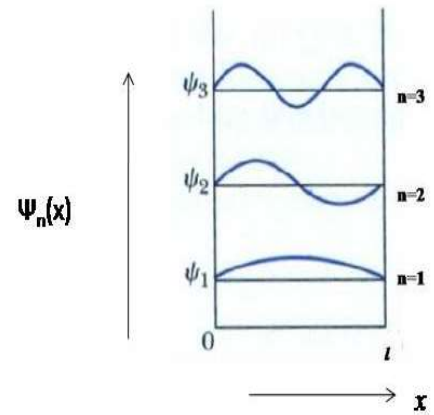
$$A^2 \int_0^l \sin^2 \frac{n\pi x}{l} dx = 1 \quad \left[\text{Since } \sin^2 x = \frac{1 - \cos 2x}{2} \right]$$

$$A^2 \int_0^l \frac{1 - \cos 2n\pi x}{2} dx = 1 \quad \left[\text{Since } \int \frac{\cos n\pi x}{l} = \frac{\sin n\pi x}{\frac{n\pi}{l}} \right]$$

$$\frac{A^2}{2} \int_0^l \frac{1 - \cos 2n\pi x}{l} dx = 1$$

$$\frac{A^2}{2} \int_0^l \frac{1 - \cos 2n\pi x}{l} dx = 1$$

$$\frac{A^2}{2} \left[\int_0^l dx - \int_0^l \frac{\cos 2n\pi x}{l} dx \right] = 1$$



$$\frac{A^2}{2} [l] - \frac{A^2}{2} \left[\frac{\sin 2n\pi l}{\frac{2n\pi}{l}} \right] = 1 \quad \longrightarrow \quad (12)$$

Since $\sin n\pi = 0$, $\sin 2n\pi$ is also $= 0$

Equation (12) can be written as

$$\frac{A^2}{2} [l] = 1$$

$$A^2 = \frac{2}{l}$$

$$\boxed{A = \sqrt{\frac{2}{l}}} \quad \longrightarrow \quad (13)$$

Substituting the values of 'A' in equation (10),

$$\Psi_n(x) = \sqrt{\frac{2}{l}} \sin \frac{n\pi x}{l} \quad \longrightarrow \quad (14)$$

Each value of wave function is known as Eigen function. The various Eigen functions of an electron enclosed in a 1D box as shown in fig.

Case 1: If $n=1$, then $\Psi_1(x)$ has two nodes at $x=0$ and $x=l$.

Case 2: If $n=2$, then $\Psi_2(x)$ has two nodes at $x=0$ and $x=\frac{l}{2}$ and $x=l$.

Case 3: If $n=3$, then $\Psi_3(x)$ has three nodes at $x=0$ and $x=\frac{l}{3}$ and $x=\frac{2l}{3}$ and $x=l$.

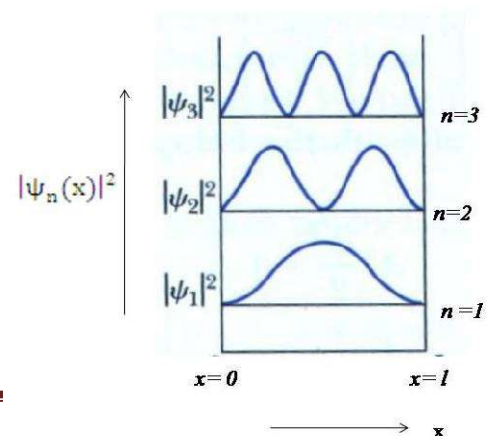
Therefore $\Psi_n(x)$ has $(n+1)$ nodes.

Probability of the location of the particle:

The probability of finding a particle over a small distance dx at x is given by

$$P(x) dx = |\Psi_n(x)|^2 dx$$

$$P(x) dx = \frac{2}{l} \sin^2 \frac{n\pi x}{l}$$



Thus, the probability density for one dimensional motion is $P(x) = \frac{2}{l} \sin^2 \frac{n\pi x}{l}$

The probability density is maximum when $\frac{n\pi x}{l} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$

$$x = \frac{l}{2n}, \frac{3l}{2n}, \frac{5l}{2n} \dots\dots$$

Case 1: If $n=1$, then the probability of position of the particle is at $x = \frac{l}{2}$

Case 2: If $n=2$ then the probability of position of the particle is at $x = \frac{l}{4}$ and $\frac{3l}{4}$.

Case 3: If $n=3$, then the probability of position of the particle is at $x = \frac{l}{6}$ and $\frac{3l}{6}$ and $\frac{5l}{6}$. these positions are shown in fig.

Energy of the Particle:

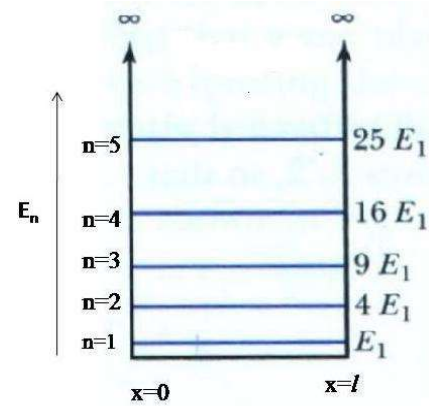
From the equations (3) & (9)

$$k^2 = \frac{8\pi^2 m E}{h^2} \rightarrow (3)$$

$$k^2 = \frac{n^2 \pi^2}{l^2} \rightarrow (9)$$

$$\frac{8\pi^2 m E}{h^2} = \frac{n^2 \pi^2}{l^2}$$

$$\boxed{E_n = \frac{n^2 h^2}{8ml^2}} \rightarrow (15)$$



This is an expression for the energy of the particle.

Each value of E_n is known as Eigen value.

The various Eigen values of an electron enclosed in a 1D box as shown in fig

Case 1: If $n=1$, then $E_1 = \frac{h^2}{8ml^2}$ This is known as zero point energy of the system.

Case 2: If $n=2$, then $E_2 = \frac{4h^2}{8ml^2} = 4E_1$

Case 3: If $n=3$, then $E_3 = \frac{9h^2}{8ml^2} = 9E_1$ and so on. Therefore, energy levels of electron are discrete.